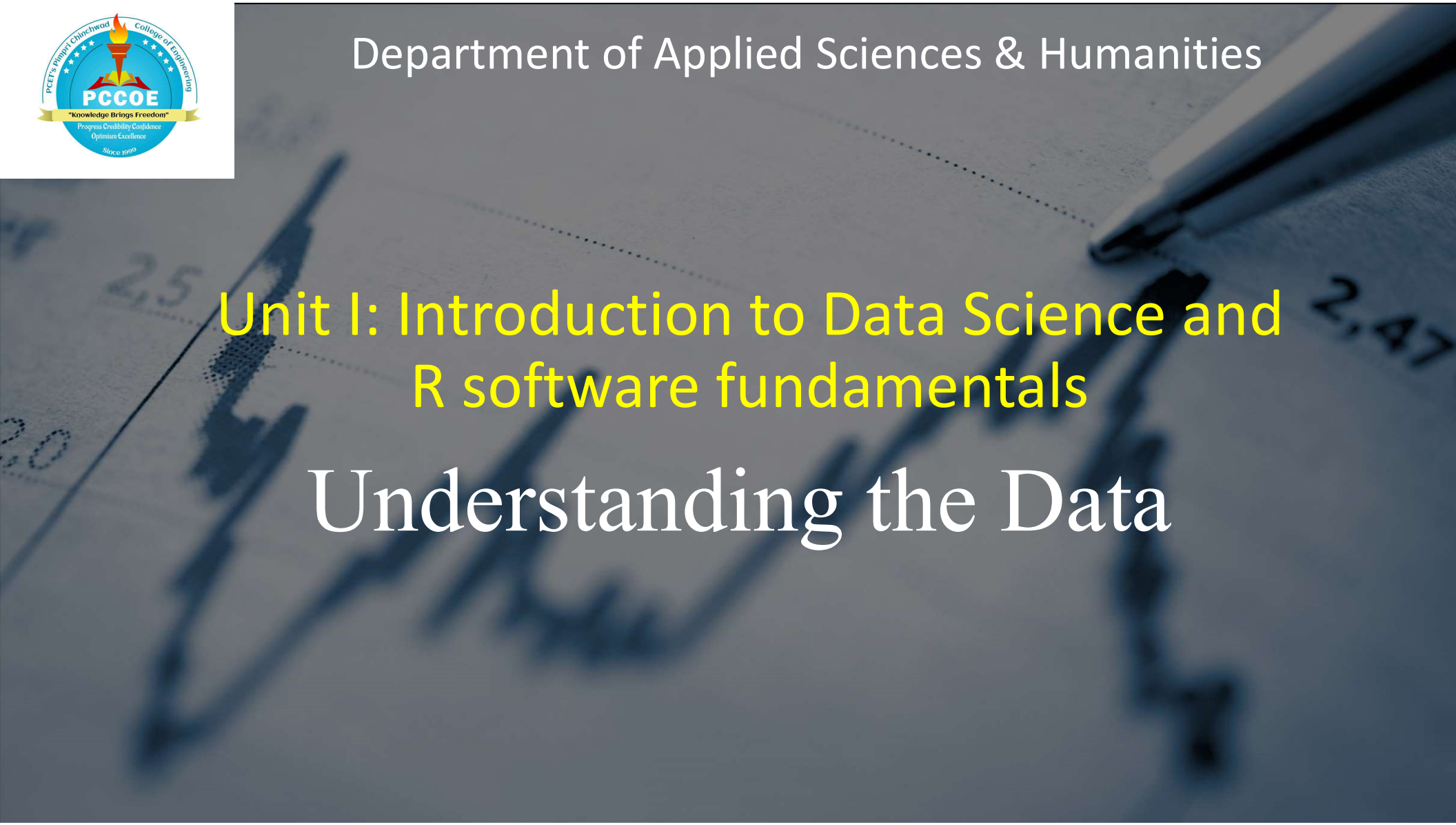


Department of Applied Sciences & Humanities

Unit I: Introduction to Data Science and R software fundamentals

Understanding the Data

Situation Analysis

Every day we hear:

- Data is Big
- Data is Open
- Data is Linked

We almost never hear about:

- What Data actually is
- How its Represented
- How its Accessed, Shared, and Integrated

What is Data?

How we express Observation in reusable form.



What is Data?

Data is any set of characters that is gathered and translated for some purpose, usually analysis.

If data is not put into context, it doesn't do anything to a human or computer.

In a computer's storage, digital data is a series of bits (binary digits) that have the value one or zero. Data is processed by the CPU, which uses logical operations to produce new data (output) from source data (input).

Data can be in form of

- Single character
- Boolean (true or false)
- Text (string)
- Number (integer or floating-point)
- Picture
- Sound
- Video
- Signals

Difference between data and information

Data is a collection of values. Those values can be characters, numbers, or any other data type. If those values are not processed, they have little meaning to a human. **Information** is data that was processed so a human can read, understand, and use it.

The "P" in CPU stands for "processing," specifically, data processing. Processing data into information is the fundamental purpose of a computer.

Difference between data and information

Information *informs* you of something. It answers a specific question. It represents a specific truth or fact.

Data is the collection of recorded values from which information can be ascertained.

For example, consider the question, "what is the temperature outside?" Data provides the basis for an answer to that question. If the data is "25.6" and "Celsius," the answer is, "Outside, the temperature is 25.6 degrees Celsius." You must know what "temperature" is, and what "degrees Celsius" are, to process the data into information.

Examples of data and information

- **Example of Data**

Department, 411044, Applied, Ashwini, Room No., Sciences and Humanities, 6411, Vaze,Nigdi, Pccoe

In this example, the original data appears to be a set of random words and numbers, separated by commas.



- **Example of Information**

Ashwini Vaze,

Room No. 6411,

Department of Applied sciences and Humanities,

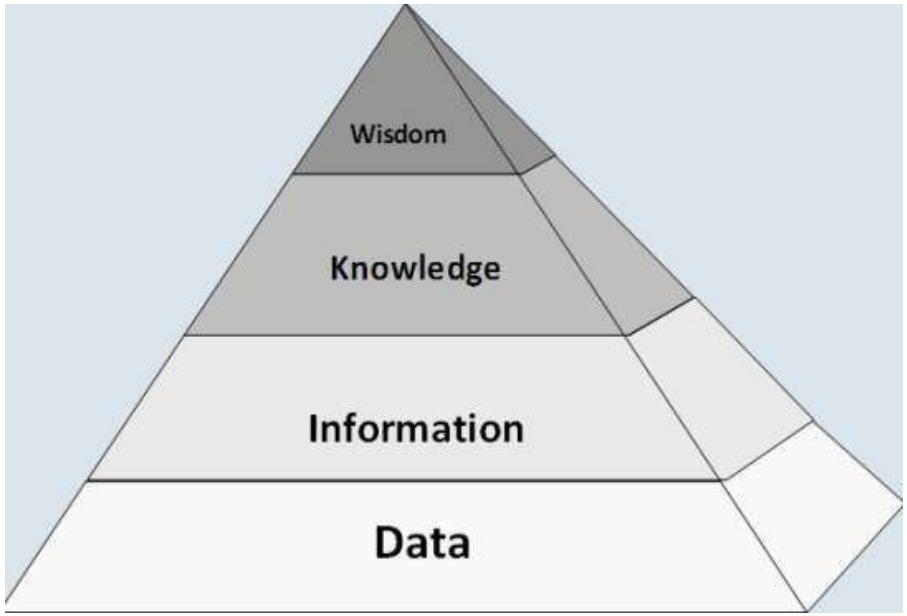
PCCoE, Nigdi, 411044

In this information, the original data was interpreted, organized, and formatted according to predefined parameters.

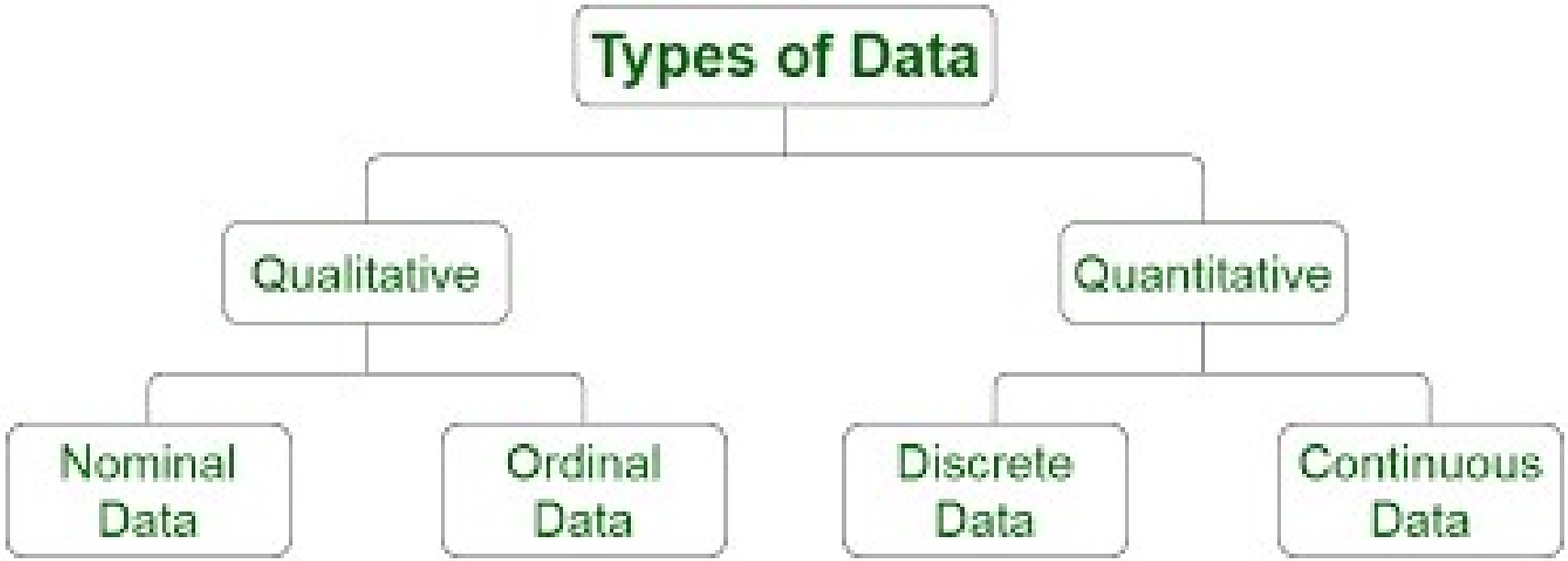
Now the meaning of the above data is clear: it is the contact information for a person named Ashwini Vaze.

Why is Data Important?

It's basis for Information, Knowledge, and Wisdom



Types of Data



Qualitative (Categorical) data

- This type of data expressed in non-numerical characters.
- It is expressed as images, videos, text documents, or audio. This type of data can't be measured or counted.
- It is used to determine how people feel about something – it's about people's feelings, motivations, opinions, and perceptions and involves bias.
- It is descriptive and aims to answer questions such as 'Why', 'How', and 'What'.
- Qualitative data is gathered from observations, surveys, or user interviews.

Qualitative or categorical data can be **Ordinal, or Nominal**

- **Ordinal** means ordered on degrees of, or ranked scales like Distinction, First class , Second class etc.
- **Nominal** data can be classified as mutually exclusive categories with in a variable. Ex: gender [male or female]

NOMINAL DATA

Nominal data divides variables into mutually exclusive, labeled categories.

Examples

Eye color



Blue Brown Green

Smartphone



iPhone Samsung Moto

Transport



Bus Train Car

How is nominal data analyzed?

- **Descriptive statistics**: Frequency distribution and mode
- **Non-parametric** statistical tests

ORDINAL DATA

Ordinal data classifies variables into categories which have a natural order or rank.

Examples

School grades



Education level



Seniority level



How is ordinal data analyzed?

- **Descriptive statistics**: Frequency distribution, mode, median, and range
- **Non-parametric** statistical tests

Quantitative (Numerical) data

- This data is expressed in numerical characters. This type of data is countable, measurable, and comparable.
- It is about amounts of numbers and involves things such as quantity and the average of numbers. It aims to answer questions such as ‘How much, ‘How many’, ‘How often’, and 'How long’.
- The act of collecting, analyzing, and interpreting quantitative data is known as performing statistical analysis.
- Statistical analysis helps uncover underlying patterns and trends in data.

Quantitative data is of the two type **Discrete** and **Continuous**.

Discrete data is information that is collected and can be counted. This type of data only has a certain number of values. Typically, it involves integers—whole numbers, rather than fractions.

Continuous data is data that can take any value and can be infinitely broken down into smaller parts. It differs from discrete data in that it can change its value. Continuous data can also fluctuate over time.

TYPES OF DATA

Qualitative Data

Names as John, Patricia, Mary…



Smell e.g. aromatic, buttery,....



Colors e.g. green, white, blue....



Quantitative Data

Scores on exams e.g. 85, 67, 90 ...

1 2 3

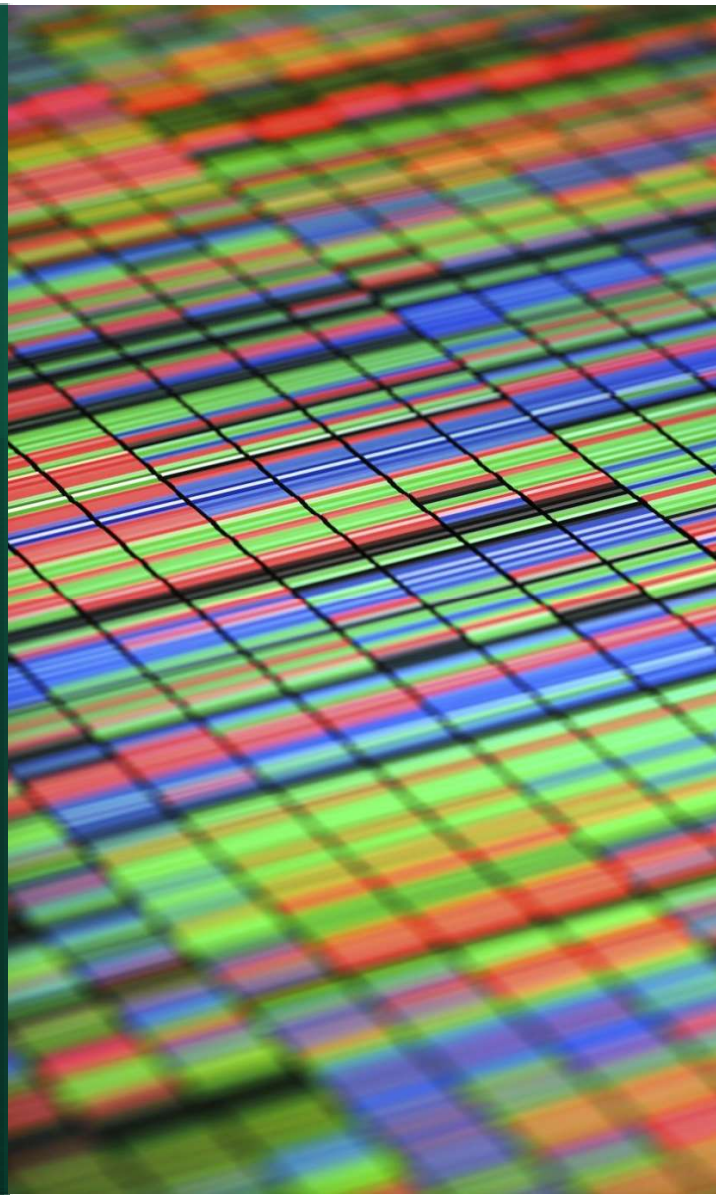
The weight of a person or a subject



Your shoe size



Data Collection



Data Collection

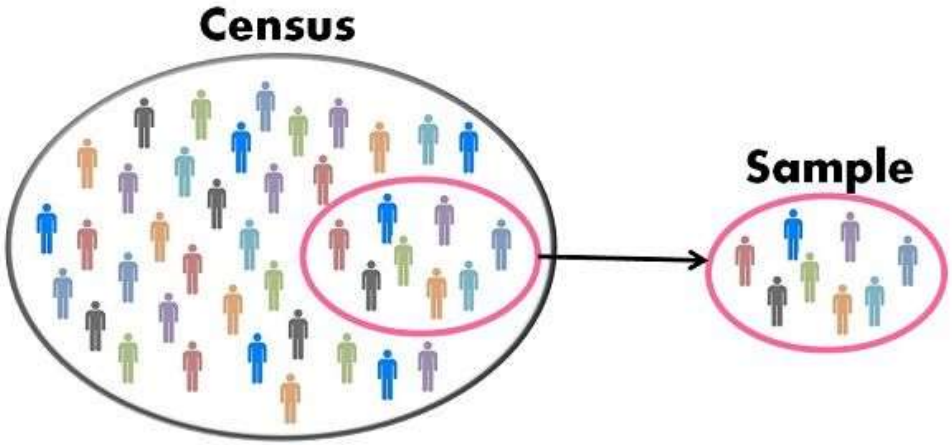
Data collection is the process of collecting and evaluating information or data from multiple sources to find answers to research problems, answer questions, evaluate outcomes, and forecast trends and probabilities.

It is an essential phase in all types of research, analysis, and decision-making, including that done in the social sciences, business, and healthcare.

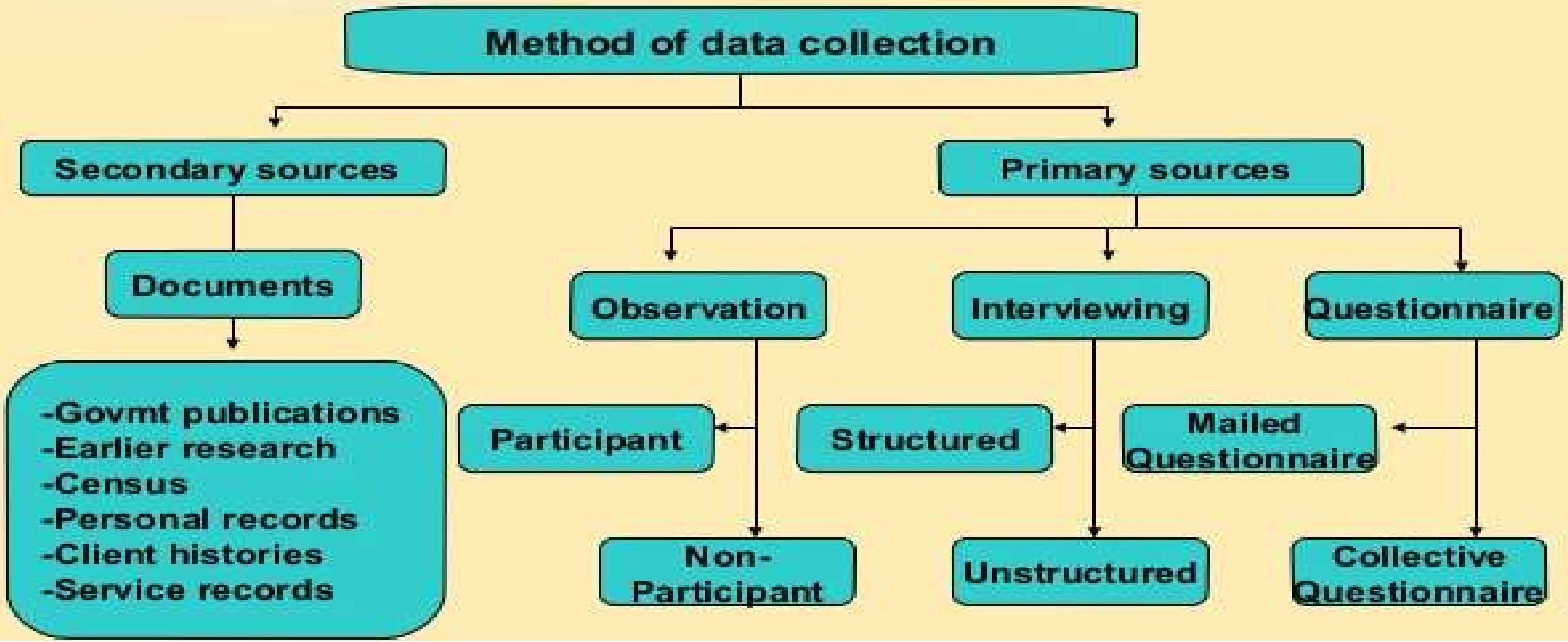
Accurate data collection is necessary to make informed business decisions, ensure quality assurance, and keep research integrity.

Data collection tools

- Data could be collected by:
  - conducting a census
  - conducting a sample survey
  - online/web Surveys
  - use of administrative records
  - conducting experiments
  - observations
  - review of secondary sources



Methods of Data Collection



Primary Data Sources



Experiments
Interviews
Surveys
Examinations
Observations

Key Factors to take care for data collection

- Data should not be contaminated by poor measurement or errors in procedure.
- Eliminate confounding variables from study or minimize effects on variables.
- Representativeness: Does your sample represent the population you are studying? Must use random sample techniques.
- Freedom from Bias
- Freedom from Confounding
- Control of Extraneous Variables
- Statistical Precision to Test Hypothesis

QUALITATIVE METHODS

The qualitative methods most commonly used in evaluation can be classified in three broad categories:

- In-depth interview
- Observation methods
  - Participant observation
  - Direct observation
- Document review

These methods are characterized by the following attributes:

- They tend to be open-ended and have less structured protocols
- They rely more heavily on interactive interviews

QUANTITATIVE METHODS

Typical quantitative data gathering strategies include:

- Experiments/clinical trials.
- Observing and recording well-defined events (e.g., counting the number of patients waiting in emergency at specified times of the day).
  - Obtaining relevant data from management information systems.
  - Administering surveys with closed-ended questions

These methods are characterized by the following attributes:

- Objectivity for the information
- Generalizability
- Precision
- Comparison should be allowed

STEPS IN DATA COLLECTION

STEP 1

- **Formulate the problem**
  - Develop objectives of data collection
  - Plan, human resource, logistics, scheduling, budgeting
  - Discuss with stakeholders

STEP 2

- **Determine sources of information**
  - Define approach to data collection
  - Identify concepts, definitions and classifications to be used

STEPS OF DATA COLLECTION …Contd

STEP 3:

- **Determine techniques of data collection**
  - Determine best approach to data collection

STEP 4:

- **Prepare data collection forms**
    - Design data collection forms

STEP 5:

- **Pretest data collection instruments**
  - Collect some information to refine the questionnaire/ data collection form.
  - Determine feasibility of obtaining data

STEPS OF DATA COLLECTION …Contd

STEP 6:

- **Finalise data collection forms**
  - Discuss final questionnaire/form with stakeholders and reproduce questionnaires/forms

STEP 7:

- **Collect data**
  - Put in place a team of data collectors/ fieldworkers
  - Train data collectors

Secondary Data Sources



- Government publications
- Research Organizations
- Government sources
- Educational Institutes
- Journals, etc

From where we get “Secondary Data”

1. Kaggle website
https://www.kaggle.com/datasets/prasad22/healthcare-dataset

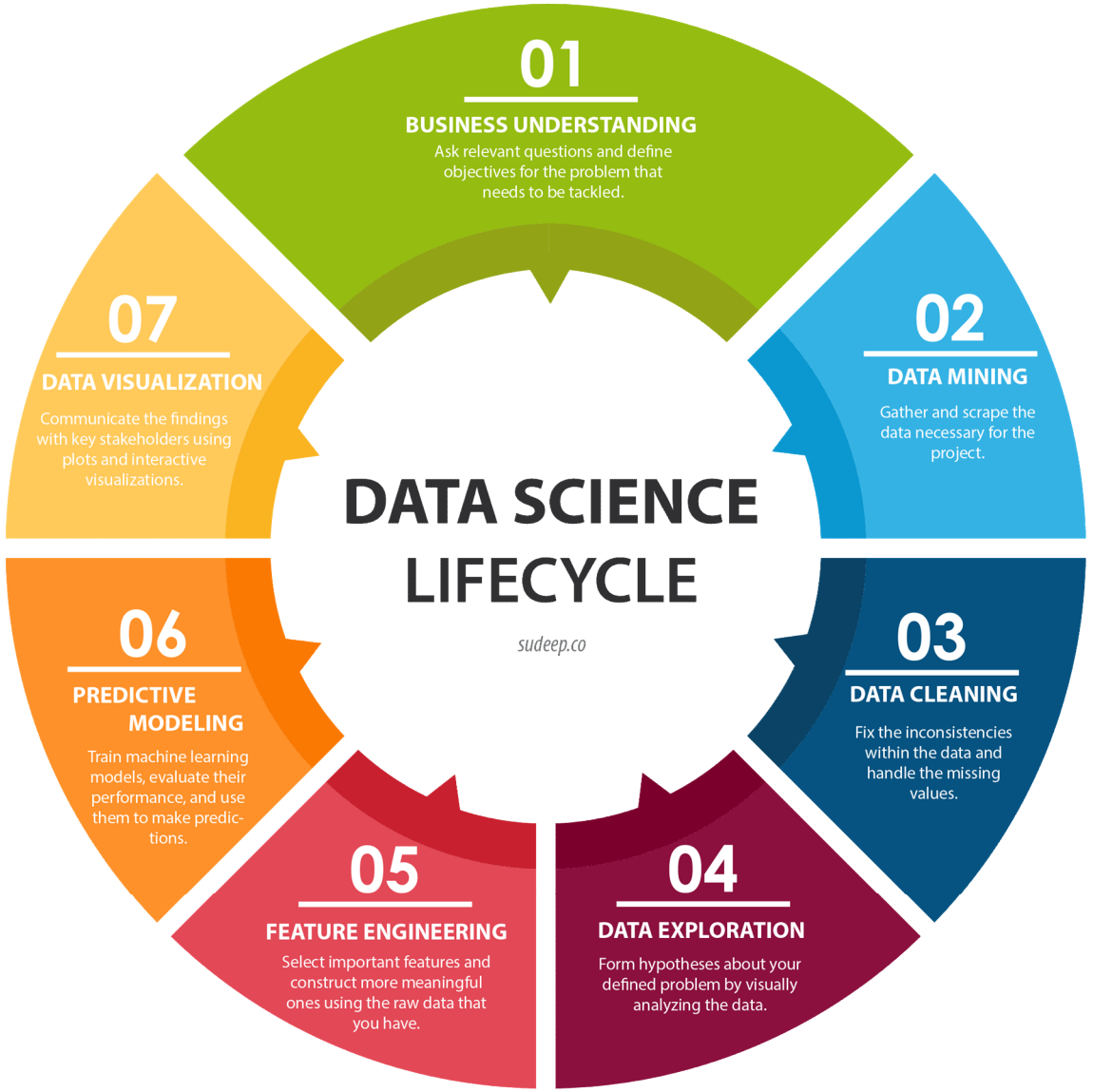
2. Government of India Ministry of Statistics and Programme Implementation
https://www.mospi.gov.in/

https://www.mospi.gov.in/publication/women-men-india-2022

https://statsofindia.in/

Data Analysis

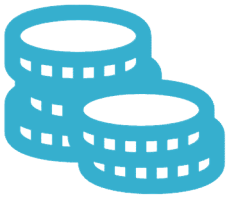
Data Analysis is the process of converting data into useful information



Advantages of Data Analysis



Informed decision-making



Reduce costs



Target customer's better

Additionally, advantages can be seen for **Operational Efficiency, Competitive advantage, Risk Management, Personalization, Innovation**

Areas of Application

- Information security
- Customer relationship management
- Finance
- Fraud detection
- Real-time credit scoring
- Logistics
- Marketing
- E-Commerce
- Academics
- Research
- Healthcare
- Cybersecurity
- Stock Market
- Banking

EXAMPLES

- Let’s have a quick and simple example. This example can be applied to restaurants or food delivery services.
- Sometimes a mail pops up in your inbox with the subject title “We missed you!” or “We have been wondering why you haven’t been in for a while” if you haven’t visited or used their services for more than a month.
- These emails may claim special offers or discounts to attract customers. In fact, these restaurants or food delivery services actually use guest management software for tracking the ordering habits of their customers and guests.
- These software tools provide email campaigning to targeted customers and help in improving business performance.

Example

- Another common example is E-commerce. When you visit these websites and start searching for a particular product, in no time you can see various recommendations of that product from various brands.
- Here the business intelligence should be in the right place i.e., the site should list the recommendations of earphones but not mobile phones if you search for earphones.
- Even though the use of these data analytics is deployed as a standard in almost every business, **there are challenges regarding the data quality concerns, reporting times, software architecture, and the actual way of data usage.** Choosing the right business approaches and data strategies can help in overcoming these issues.

Questions

- What is data? Explain the types of data with examples
- What is Data Analysis? Explain in detail.
- Elaborate process of Data Analysis
- State the advantages of Data Analysis.
- Elaborate any 5 areas of application of Data Analysis.